原版PT的MFVI推导

符号约定

d: `dim_z`

m: `dim_g`

n: `seq_len`

势函数

$\phi_u(Z_i = a) = \exp(S_{w_i,a})$
$\phi_t(H_i = k, Z_i = a, Z_j = b) = \exp(T_{a,b})(H_i = j, \text{否则为}1)$
$\phi_b(Z_i = a, G_i = g) = \exp(B_{g,a})$

联合概率分布

给定序列w（坐标为$i \in 1 \sim n$）,给定各位置i的Z_i, G_i, H_i，此取值概率为：$P(Z, H, G|w) = \text{Normalization}\{\Pi_{i=1}^n \phi_u(Z_i) * \Pi_{i=1}^n \phi_b(Z_i, G_i) * \Pi_{c=1}^h \Pi_{i=1}^n \Pi_{j=1}^n \phi_t(H_i^{(c)}, Z_i, Z_j)\}$

MFVI

变分自由能函数：$F(Q) = E_Q[-\log P(Z, H, G|w)] - H(Q)$
$= -\sum_{i=1}^n (\sum_{a=1}^d Q_i(Z_i = a) S_{w_i,a})$
$-\sum_{c=1}^C \sum_{i=1}^n (\sum_{j \neq i} \sum_{a=1}^d \sum_{b=1}^d Q_{ic}(H_i = j) Q_i(Z_i = a) Q_j(Z_j = b) T_{a,b}^{(c)})$
$-\sum_{i=1}^n (\sum_{a=1}^d \sum_{g=1}^m Q_i(Z_i = a) Q_i^G(G_i = g) B_{g,a})$
$+H(Q)$

MFVI: 求$Q_i(a)$分布使得$F(Q)$取得最小值，且$\sum_a Q_i(a) = 1$。以此构建Lagrange函数，使其对$Q_i(a)$偏导为0，解出一次迭代后的$Q_i(a)$。

Lagrange:$L = F(Q) + \lambda(\sum_z Q_i(a) - 1)$
偏导为0$\Rightarrow \nabla E + \ln Q_i(a) + 1 + \lambda = 0$ (其中∇E为$E_Q[-\log P(Z, H, G|w)]$对$Q_i(a)$的偏导，$\ln Q_i(a) + 1$为$H(Q)$对$Q_i(a)$的偏导)

$\nabla E_{Q_{ic}(j)}$的求解与$Q_{ic}(j)$的更新：

$\frac{\partial E}{\partial Q_{ic}(j)} = \sum_a \sum_b Q_i(a) Q_j(b) T_{a,b}(c)$
$= \sum_a \sum_b Q_i(a) Q_j(b) (\sum_l U_{a,l} V_{b,l})$

$= \sum_l (\sum_a Q_i(a) U_{a,l})(\sum_b Q_j(b) V_{b,l})$
令$\sum_a Q_i(a) U_{a,l} = q_{i,l}, \sum_b Q_j(b) V_{b,l} = k_{j,l}$
则$\nabla E_{Q_{ic}(j)} = \sum_{l=1}^{r} q_{i,l} k_{j,l} = -(\vec{q_i} \vec{k_j^T})$

由$Q \propto \exp(-\nabla E)$得：$Q_{ic}(j) \propto \exp(\vec{q_i} \vec{k_j^T})$

$\nabla E_{Q_i(a)}$的求解与$Q_i(a)$的更新：

对F,只考虑含$Q_i(a)$的项：
$F_i(Q) = -Q_i(a) S_{w_i,a}$
$- \sum_c \sum_{j \neq i} Q_{ic}(j) \sum_b Q_i(a) Q_j(b) T_{ab}^{(c)}$
$- \sum_c \sum_{k \neq i} Q_{kc}(i) \sum_b Q_k(b) Q_i(a) T_{ba}^{(c)}$
$- \sum_g Q_i(a) Q_i^G(g) B_{g,a}$

$\nabla_{unary} = -S_{w_i,a}$
$\nabla_{child} = - \sum_c \sum_{j \neq i} Q_{ic}(j) (\sum_b Q_j(b) T_{a,b}^{(c)})$
$\nabla_{head} = - \sum_c \sum_{k \neq i} Q_{kc}(i) (\sum_b Q_k(b) T_{b,a}^{(c)})$
$\nabla_{binary} = - \sum_g Q_i^G(g) G_{g,a}$
$\therefore \nabla E = \nabla_{unary} + \nabla_{child} + \nabla_{head} + \nabla_{binary}$
代入Lagrange函数，令其偏导为0$\Rightarrow \ln Q_i(a) = -\nabla E - (1 + \lambda)$
$\Rightarrow Q_i(a) \propto \exp(-\nabla E)$
$\Rightarrow Q_i(a) = \mathrm{Softmax}(-\nabla E)$

即：$\mathrm{Logits}_i(a) = -\nabla E$
$= S_{w_i,a} + \sum_c \sum_{j \neq i} Q_{ic}(j) (\sum_b Q_j(b) T_{a,b}^{(c)}) + \sum_c \sum_{k \neq i} Q_{kc}(i) (\sum_b Q_k(b) T_{b,a}^{(c)}) +$
$\sum_g Q_i^G(g) B_{g,a}$

考虑T矩阵分解：

$\sum_{j \neq i} Q_{ic}(j) (\sum_b Q_j(b) T_{a,b}^{(c)}) = \sum_j \mathrm{Attn}_{ij}(Q_j T) = \sum_j \mathrm{Attn}_{ij}(Q_j V) U^T$
$\sum_{k \neq i} Q_{kc}(i) (\sum_b Q_k(b) T_{b,a}^{(c)}) = \sum_k \mathrm{Attn}_{ki}(Q_k T) = \sum_k \mathrm{Attn}_{ki}(Q_k U) V^T$

相应代码： `qz_new = Softmax(Input_S + (Attn_Child + Attn_Head) + FFN_Output)`

muP版 PT的MFVI推导

势函数 (muP Scaled)

$\phi_u(Z_i = a) = \exp(S_{w_i,a})$
$\phi_t(H_i = k, Z_i = a, Z_j = b) = \exp(\frac{d^2}{r}T_{a,b})(H_i = j, \text{否则为}1)$
$\phi_b(Z_i = a, G_i = g) = \exp(d \cdot mB_{g,a})$

联合概率分布

给定序列w（坐标为$i \in 1 \sim n$）,给定各位置i的Z_i, G_i, H_i，此取值概率为：$P(Z, H, G|w) = \text{Normalization}\{\Pi_{i=1}^n \phi_u(Z_i) * \Pi_{i=1}^n \phi_b(Z_i, G_i) * \Pi_{c=1}^h \Pi_{i=1}^n \Pi_{j=1}^n \phi_t(H_i^{(c)}, Z_i, Z_j)\}$

MFVI

变分自由能函数：$F(Q) = E_Q[-\log P(Z, H, G|w)] - H(Q)$
$= -\sum_{i=1}^n (\sum_{a=1}^d Q_i(Z_i = a) S_{w_i,a})$
$-\sum_{c=1}^C \sum_{i=1}^n (\sum_{j \neq i} \sum_{a=1}^d \sum_{b=1}^d Q_{ic}(H_i = j) Q_i(Z_i = a) Q_j(Z_j = b)(\frac{d^2}{r} T_{a,b}^{(c)}))$
$-\sum_{i=1}^n (\sum_{a=1}^d \sum_{g=1}^m Q_i(Z_i = a) Q_i^G(G_i = g)(d \cdot mB_{g,a}))$
$+H(Q)$

MFVI: 求$Q_i(a)$分布使得$F(Q)$取得最小值，且$\sum_a Q_i(a) = 1$。以此构建Lagrange函数，使其对$Q_i(a)$偏导为0，解出一次迭代后的$Q_i(a)$。

Lagrange:$L = F(Q) + \lambda(\sum_z Q_i(a) - 1)$
偏导为0$\Rightarrow \nabla E + \ln Q_i(a) + 1 + \lambda = 0$ (其中∇E为$E_Q[-\log P(Z, H, G|w)]$对$Q_i(a)$的偏导，$\ln Q_i(a) + 1$为$H(Q)$对$Q_i(a)$的偏导)

$\nabla E_{Q_{ic}(j)}$的求解与$Q_{ic}(j)$的更新：

$\frac{\partial E}{\partial Q_{ic}(j)} = \sum_a \sum_b Q_i(a) Q_j(b)(\frac{d^2}{r} T_{a,b}(c))$
引入代码变量 $q^{\text{code}} = dQ$，则 $Q = q^{\text{code}}/d$:
$= \frac{d^2}{r} \sum_a \sum_b \frac{q_i^{\text{code}}(a)}{d} \frac{q_j^{\text{code}}(b)}{d} T_{a,b}^{(c)}$
$= \frac{1}{r} \sum_a \sum_b q_i^{\text{code}}(a) q_j^{\text{code}}(b)(\sum_l U_{a,l} V_{b,l})$
$= \frac{1}{r} \sum_l (\sum_a q_i^{\text{code}}(a) U_{a,l})(\sum_b q_j^{\text{code}}(b) V_{b,l})$
令$\sum_a q_i^{\text{code}}(a) U_{a,l} = q_{i,l}^{\text{proj}}$,$\sum_b q_j^{\text{code}}(b) V_{b,l} = k_{j,l}^{\text{proj}}$
则$\nabla E_{Q_{ic}(j)} = \frac{1}{r} \sum_{l=1}^r q_{i,l}^{\text{proj}} k_{j,l}^{\text{proj}} = -\frac{1}{r}(\vec{q_i^{\text{proj}}} \vec{k_j^{\text{proj}T}})$

由$Q \propto \exp(-\nabla E)$得：$Q_{ic}(j) \propto \exp(\frac{1}{r} \vec{q_i^{\text{proj}}} \vec{k_j^{\text{proj}T}})$

$\nabla E_{Q_i(a)}$的求解与$Q_i(a)$的更新：

对F,只考虑含$Q_i(a)$的项：
$F_i(Q) = -Q_i(a) S_{w_i,a}$
$- \sum_c \sum_{j \neq i} Q_{ic}(j) \sum_b Q_i(a) Q_j(b) (\frac{d^2}{r} T_{ab}^{(c)})$
$- \sum_c \sum_{k \neq i} Q_{kc}(i) \sum_b Q_k(b) Q_i(a) (\frac{d^2}{r} T_{ba}^{(c)})$
$- \sum_g Q_i(a) Q_i^G(g) (d \cdot m B_{g,a})$

引入代码变量 $q^{\text{code}} = dQ$ 及 $q^{G,\text{code}} = mQ^G$：

$\nabla_{unary} = -S_{w_i,a}$
$\nabla_{child} = -\sum_c \sum_{j \neq i} Q_{ic}(j) (\sum_b \frac{q_j^{\text{code}}(b)}{d} \frac{d^2}{r} T_{a,b}^{(c)}) = -\frac{d}{r} \sum_c \sum_{j \neq i} Q_{ic}(j) (\sum_b q_j^{\text{code}}(b) T_{a,b}^{(c)})$
$\nabla_{head} = -\sum_c \sum_{k \neq i} Q_{kc}(i) (\sum_b \frac{q_k^{\text{code}}(b)}{d} \frac{d^2}{r} T_{b,a}^{(c)}) = -\frac{d}{r} \sum_c \sum_{k \neq i} Q_{kc}(i) (\sum_b q_k^{\text{code}}(b) T_{b,a}^{(c)})$
$\nabla_{binary} = -\sum_g \frac{q_i^{G,\text{code}}(g)}{m} (d \cdot m B_{g,a}) = -d \sum_g q_i^{G,\text{code}}(g) B_{g,a}$

$\therefore \nabla E = \nabla_{unary} + \nabla_{child} + \nabla_{head} + \nabla_{binary}$
代入Lagrange函数，令其偏导为0$\Rightarrow \ln Q_i(a) = -\nabla E - (1 + \lambda)$
$\Rightarrow Q_i(a) \propto \exp(-\nabla E)$
$\Rightarrow Q_i(a) = \text{Softmax}(-\nabla E)$

即：$\text{Logits}_i(a) = -\nabla E$
$= S_{w_i,a} + \frac{d}{r} \sum_c \sum_{j \neq i} Q_{ic}(j) (\sum_b q_j^{\text{code}}(b) T_{a,b}^{(c)}) + \frac{d}{r} \sum_c \sum_{k \neq i} Q_{kc}(i) (\sum_b q_k^{\text{code}}(b) T_{b,a}^{(c)}) + d \sum_g q_i^{G,\text{code}}(g) B_{g,a}$

考虑T矩阵分解及代码对应（常数系数 d 被吸收到 Scaling 或作为能量补偿体现）：

$\text{Term}_{child} \propto \sum_j \text{Attn}_{ij} (q_j^{\text{code}} V) U^T$
$\text{Term}_{head} \propto \sum_k \text{Attn}_{ki} (q_k^{\text{code}} U) V^T$
$\text{Term}_{binary} \propto q^{G,\text{code}} B$

相应代码：

```
message_F = (qz_u @ qz_v.T) / ternary_rank
qz_new = Softmax(Input_S + (Attn_Child + Attn_Head) + FFN_Output)
```